

Brush and Dry: A Posttreatment-Free Simple Process to Pattern PEDOT:PSS Films with Enhanced Thermoelectric Performance via Shear Flow-Induced Morphology Tuning

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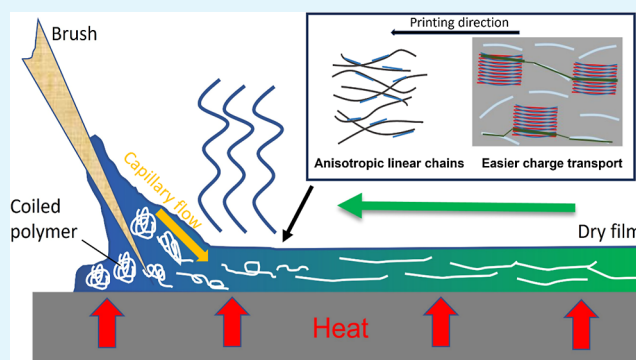
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ABSTRACT: Brush printing is a promising meniscus-guided technique to facilitate the transition from optimized organic thermoelectric materials to functional printed thermoelectric generators (TEGs) due to its dual advantage: (1) precise control of solvent evaporation to tune film morphology for enhanced charge transport and (2) patterning capabilities. This study systematically investigates the relationship between brush-induced morphology and thermoelectric performance in PEDOT:PSS films. We observed molecular chain extension, anisotropic alignment, and improved crystalline packing under optimal brushing conditions. This optimized morphology led to a 3-fold increase in electrical conductivity, up to 1900 S cm^{-1} , and a high power factor of $170 \mu\text{W m}^{-1} \text{ K}^{-2}$, without any wet posttreatment steps. This work demonstrates that brush printing offers a simple and scalable process route for organic thermoelectrics, effectively bridging the gap between high-performance materials and their practical applications in TEGs.

KEYWORDS: organic thermoelectrics, PEDOT:PSS, meniscus-guided printing, brush printing, molecular alignment



INTRODUCTION

Self-powered technologies have expanded rapidly in recent years, spurred by the growing demand for Internet of Things (IoT) devices, which require extended lifespans. Thermoelectric materials, capable of directly converting thermal energy into electricity, present a promising alternative for sustainable energy harvesting. Thermoelectric efficiency is quantified by the dimensionless figure of merit, $zT = \frac{S^2\sigma}{\kappa}T$, where S is the Seebeck coefficient, σ is electrical conductivity, κ is thermal conductivity, and T is the absolute temperature. The power factor $PF = S^2\sigma$ is used as a performance indicator for thin films for which thermal conductivity is hard to measure or irrelevant if the substrate acts as a thermal shunt. Among thermoelectric materials, organic thermoelectrics (OTEs) have attracted attention due to their low thermal conductivity, mechanical flexibility, cost-effectiveness, and versatile processing methods. Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is one of the most extensively studied p-type polymer for OTEs because it offers ambient stability, semimetallic conductivity, and easy processing.² However, its power factor remains lower than that of inorganic thermoelectrics. To enhance the performance of PEDOT:PSS, researchers have explored various posttreatment techniques using acids (H_2SO_4) and polar solvents like dimethyl sulfoxide

(DMSO), ethylene glycol (EG), and methanol (MeOH) to (1) induce phase transformation from PEDOT regions insulated by surrounding PSS to well-interconnected PEDOT fibrous structures and (2) remove excess of the insulating PSS. These treatments have consistently increased electrical conductivity to values above 3000 S cm^{-1} ,^{3–5} and by adjusting the oxidation level authors have further improved the Seebeck coefficient to optimize the power factor to values above $300 \mu\text{W m}^{-1} \text{ K}^{-2}$.^{6–8} Notably, Hinckley et al. combined solution-shearing of PEDOT:PSS thin films and directional solvent posttreatments, achieving a record-high electrical conductivity of 8500 S/cm and power factor of $800 \mu\text{W m}^{-1} \text{ K}^{-2}$.⁹ The processes reported leading to high conducting PEDOT:PSS, unfortunately, do not take into account practical aspects for the development of thermoelectric generators (TEGs), such as scalability and patternability. These aspects are crucial because a TEG requires multiple legs interconnected in series.

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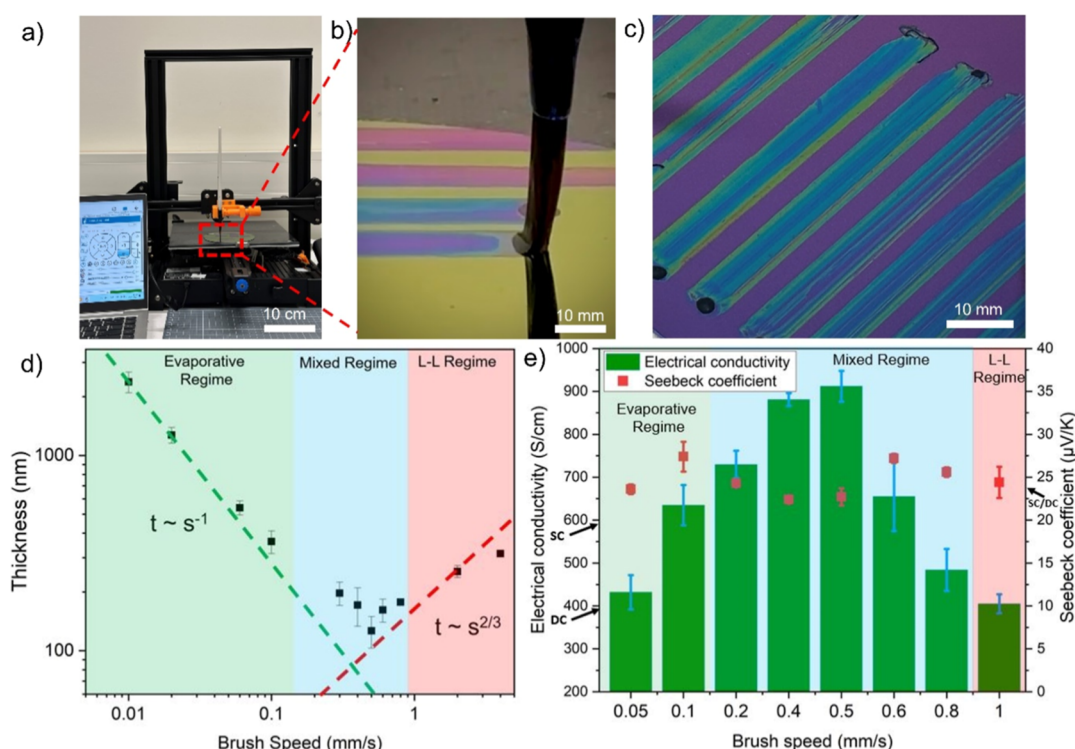


Figure 1. (a) Brush printing setup. (b) A meniscus forms at the brush tip during the printing process. (c) Brush-printed PEDOT:PSS lines. (d) Measured film thickness t as a function of brush speed s (log–log scale). The dashed lines correspond to the theoretical models for the evaporative and Landau–Levich (L–L) regimes defined for other meniscus-guided techniques. (e) Electrical conductivity and Seebeck coefficient for different brush speeds. A spin-coated (SC) film and a drop-casted (DC) film are used as references.

Moreover, the reported posttreatments leading to highly conducting PEDOT:PSS are based on wet processes that remove the PSS excess at the interface with the substrate, provoking delamination of the functional PEDOT:PSS layer. This issue, which is highly overlooked in the literature, diminishes the fabrication yield of the wet posttreated PEDOT:PSS films. While the impact of this yield reduction can be comfortably assimilated by works focusing on material properties of blank films, it is a stumbling block for TEGs, for which the failure of one leg supposes the failure of the whole device (with all legs being connected in series). Hence, a purely material-focused fabrication approach ignoring the practical implementation of TEGs hinders the transition from promising thermoelectric materials to real-world applicable devices.¹⁰ Some authors tackled the issue of patterning by using printing electronics techniques to fabricate in-plane TEGs.^{11–13} Unsurprisingly, in all of the mentioned cases the authors did not perform postprocessing dipping, which suggests that the delamination problem is still standing. As a result, the conductivity reported for those patterned TE devices was limited to a max 700 S cm^{-1} , far from the record few thousands S cm^{-1} reported for heavily posttreated PEDOT:PSS films.¹¹ Defining a strategy to boost the performance of printed PEDOT:PSS lines without subjecting them to wet postprocessing will strongly benefit the fabrication of low-cost and scalable high-performance thin-film organic TEGs.

In our work, we applied brush printing as a simple process to enhance PEDOT:PSS performance by tuning the film morphology, while enabling versatile patterning of TEG legs. Like bar coating and solution shearing,^{14,15} brush printing is a meniscus-based solution-processing technique that allows

precise thickness and morphology control by varying the shear flow via the deposition speed.¹⁶ Brush printing has been used in several semiconducting polymers to boost their mobility via shear-induced aggregation and molecular alignment.¹⁷ In PEDOT:PSS, brush printing has also induced an increase in conductivity associated with enhanced molecular order, although a detailed description of the processing-morphology-performance relationship is still missing. Contrary to the other meniscus-based techniques, brush printing allows patterning. Conveniently, the relatively poor resolution of brush-printing (dependent on the brush width) and typical high film roughness of brush-printed films are not limiting factors for TEs since the power density of planar TEGs is independent of line width,^{11,18} and the bulk transport nature of TEs makes them tolerant to surface roughness (in stark contrast to other organic devices like transistors and photovoltaics). Interestingly, although patterning and morphology control are highly desirable for fabricating TEGs, brush printing has not yet been explored in the context of TEs. We showed that our optimized brush-printed patterned lines displayed significant improvements in electrical conductivity (216%) and power factor (253%) compared to spin-coated reference blank films, reaching values as high as $1900 \pm 300 \text{ S cm}^{-1}$ and $170 \pm 37 \mu\text{W m}^{-1} \text{ K}^{-2}$, respectively. Those are the highest values reported for PEDOT:PSS without posttreatment. Indeed, this performance improvement is achieved solely through physically induced morphology without the need for any harmful or hazardous solvent. To support this claim, we provide a systematic investigation of the relationship between printing parameters and film morphology, revealing a delicate balance among shear flow-induced molecular extension, uniaxial molecular alignment, and intermolecular packing.

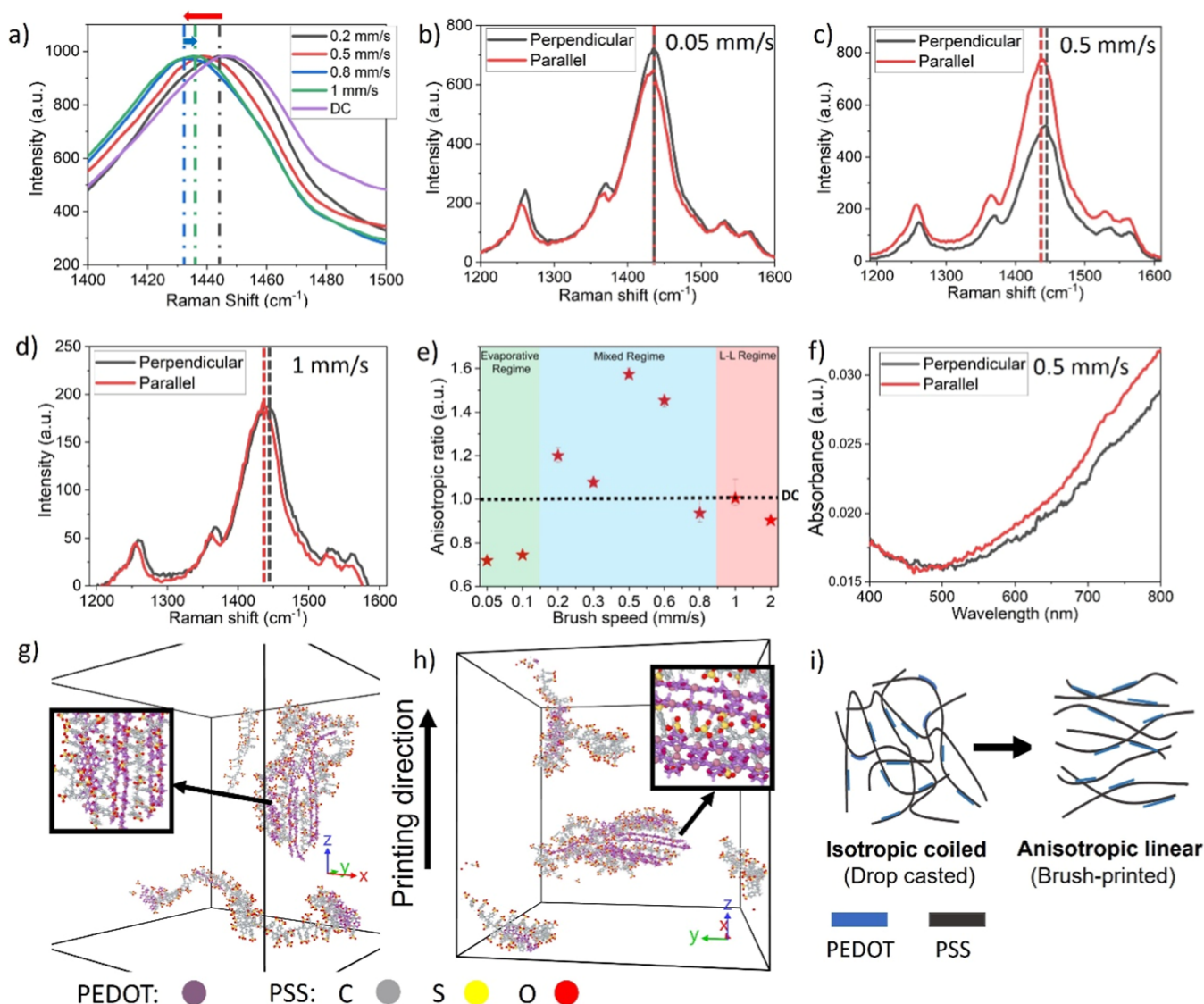


Figure 2. (a) Raman spectra of PEDOT:PSS printed at different speeds (a drop-casted film is added as a reference). Polarized Raman spectra of samples brush printed within (b) evaporative regime (0.05 mm/s); (c) mixed regime (0.5 mm/s); and (d) Landau–Levich regime (1 mm/s) probed along the directions parallel and perpendicular to brushing. (e) Anisotropy ratio of the $C_\alpha = C_\beta$ Raman peak intensity between parallel and perpendicular brushing directions versus brush speed. A drop-casted (DC) sample was used as an isotropic reference. (f) Polarized UV–Vis spectra of the film printed at 0.5 mm/s probed with the polarizer axis arranged along the directions parallel and perpendicular to brushing. (g,h) Snapshots from MD simulations illustrating the relaxation process of PEDOT strands in water at 90 °C: (g) aligned and extended chains along the brushing direction at the moment of brushing (t_0), (h) extended chains that lost alignment (t_1). Carbon atoms in PEDOT and PSS are colored in purple and gray, respectively. Sulfur atoms are yellow and oxygen atoms are red. Water and DMSO molecules have been made transparent for clarity. (i) Schematic illustration comparing the proposed morphology of PEDOT:PSS deposited by drop casting and brush-printing at 0.5 mm/s.

This work represents a significant step forward toward the cost-effective, scalable, and environmentally friendly fabrication of TEGs over large areas.

RESULTS AND DISCUSSION

The setup for brush printing and the printing meniscus is shown in Figure 1a,b (see the video of meniscus movement during brush printing in Movie S1). Our ink consisted of PEDOT:PSS aqueous solution with 5 vol % DMSO. Patterned PEDOT:PSS films were systematically brush-printed (Figure 1c) onto wafers or glass substrates using a range of brush speeds, spanning from 0.01 to 4 mm/s, while maintaining a constant substrate temperature of 100 °C and a uniform initial droplet volume of 60 μL . The study revealed the typical

distinct regimes of film formation previously reported for other meniscus-guided deposition techniques like solution shearing and bar coating (Figure 1d).^{16,19} At low brush speeds, the film thickness is inversely proportional to the brush speed, characteristic of the evaporative regime. In this regime, the solvent evaporation at the meniscus drives the film formation at the triple contact line between the meniscus, the substrate, and the air. In this regime, the meniscus is short, leading to the formation of a dry film close to the brush tip. Since the film forms directly under the influence of meniscus forces, it is easy to lock the shear-induced morphology in the solid film. Conversely, at high brush speeds, an increase in film thickness was observed with higher coating speeds, which corresponded to the Landau–Levich (L–L) regime. In this scenario, the ink is pulled out from the meniscus by viscous forces, leaving a wet

film behind the brush. Thus, film consolidation occurs far from the brush tip and out of reach of shear forces, allowing polymers to relax before film consolidation. At intermediate brush speeds, the film thickness was found to be relatively independent of the coating speed. Our empirical findings indicate that the film thickness (t) scales with speed (s) according to $t \propto s^{-0.81}$ in the evaporative regime and $t \propto s^{0.36}$ in the Landau–Levich regime. The deviation in our fitted exponents (-0.81 and 0.36) from those reported in the literature for solution-shearing or bar coating (-1 and 0.67 for evaporative and L–L regimes, respectively)¹⁹ is attributed to experimental error for the evaporative regime, and the unique meniscus shape inherent to brush printing for the L–L regime. Nevertheless, the observed trends for speed remain consistent with previous observations, providing valuable insights into the printing window to focus on for achieving optimized characteristics. It has been demonstrated for solution-sheared organic field effect transistors (OFETs) that the maximum mobility, associated with polymer chain alignment in the casting direction, was achieved in the mixed region.^{14,16} To confirm the same trend for our conducting brush-printed samples, we measured the electrical conductivity and Seebeck coefficient for the same samples used to extract the plot in Figure 1d. As shown in Figure 1e, electrical conductivity initially increased with brush speed and then started to decrease with peak performance occurring, as expected, in the mixed regime at a speed of 0.5 mm/s. In the mixed regime, the conductivity was higher than for reference spin-coated (SC) and drop-casted (DC) films with the same composition. This trend suggests a correlation between brush speed and mobility via tuning of the film morphology. In contrast, the Seebeck coefficient shows minimal variation across different printing speeds and references, indicating a stable doping level of PEDOT:PSS,¹ which is expected because brush printing is a purely physical treatment. Overall, the best thermoelectric performance ($PF = \sigma S^2$) was observed in the mixed regime.

To investigate why the mixed regime enhances electrical conductivity, we conducted a series of morphological characterization experiments on samples printed at different speeds spanning all printing regimes and with similar thicknesses. A drop-casted film, forming in the absence of any shear force, is measured as a reference. Raman spectroscopy was employed to examine the conformational changes in the PEDOT chains across the printing regimes. We assume constant doping levels; therefore, any observed peak shift in the Raman spectra reflects only morphological changes. As shown in Figure 2a, brush printing at a speed from 0.2 to 0.8 mm/s shifts the main Raman peak ($C_\alpha = C_\beta$) to lower wavenumbers compared to the drop-casted film. This peak, placed at ~ 1444 cm^{-1} , has been attributed to the conformation of the PEDOT backbone. Thus, such Raman redshift has been previously attributed to an increase in backbone extension and molecule planarity.^{20,21} This result suggests that the shear forces in the meniscus contribute to the extension of the PEDOT backbone. At 1 mm/s, corresponding to the L–L regime, the peak begins to blue-shift, indicating the onset of a reversion to a coiled backbone structure as the meniscus elongates and the stretched polymer chains have time to relax before solvent drying.

Next, we investigated the potential preferential molecular alignment along the brush printing direction within the relevant printing speed window. Polarized Raman spectra reveal material anisotropy between parallel and perpendicular

directions through intensity differences and can also detect conformational changes along those directions via a peak shift. Figure 2b–d presents the polarized Raman spectra for samples prepared under the evaporative (0.05 mm/s), mixed (0.5 mm/s), and L–L regimes (1 mm/s). To estimate the degree of alignment, the ratio of the mean peak ($C_\alpha = C_\beta$) intensity between the directions parallel and perpendicular to the printing direction was calculated (termed the anisotropic ratio). As shown in Figure 2e, the anisotropic ratio deep in the evaporative regime was below 1 (~ 0.8), indicating slight molecular alignment in the direction perpendicular to printing. This behavior has been encountered before for solution shearing at low speed.²² In the mixed regime, the anisotropic ratio exceeded 1 , reaching up to 1.6 for printing at 0.5 mm/s. This significantly stronger signal in the parallel direction compared to the perpendicular one indicates that preferential alignment of polymer chains along the brushing direction was established during printing due to shear stress, and such alignment was retained during film consolidation owing to the relatively short meniscus. The result of this process is the formation of an anisotropic solid film. Polarized UV–Vis further confirmed a preferential molecular alignment parallel to the printing direction at 0.5 mm/s (Figure 2f). A conjugated polymer displays high absorption when the direction of the molecules' backbone coincides with the polarization of the light. Thus, since higher absorption is observed when the polarizer axis is arranged along the printing direction compared to the perpendicular direction, we can conclude that there is preferential molecular backbone alignment along the printing direction. Conversely, in the L–L regime, the anisotropic ratio reverted to 1 , suggesting that the long-wet film left behind the brush facilitated the relaxation of aligned polymers before film consolidation. This film formation process resembles that of an isotropic drop-casted film. Interestingly besides the anisotropy in peak intensity, we observed that the Raman peak measured along the printing direction displayed a redshift compared to the peak measured along the perpendicular direction in Figure 2c,d, indicating that besides promoting anisotropic molecular alignment along the printing direction, brush-printing also extended the molecules preferentially in the same direction, further promoting charge transport and electrical conductivity as it was demonstrated in Figure 1e. The techniques employed to detect anisotropy for some brush-printed films reveal information at the molecular level. To further investigate whether the morphological order extends over long ranges, polarized optical microscopy (POM) was performed on the most crystalline and anisotropic sample (brush-printed at 0.5 mm/s). POM did not reveal long-range domain orientation (Figure S1). However, POM is suitable only for highly crystalline materials. The crystallinity of PEDOT:PSS might still be too low to be observable by POM, making long-range crystal orientation indiscernible. Note that similar POM results were reported by Wang et al.¹⁷ Then, we utilized atomic force microscopy (AFM) to probe the same film (Figure S2). No long-range domain orientation was observed with AFM either, which suggests that anisotropy originates mainly from localized domains.

The conductivity ratio along the directions parallel and perpendicular to printing is detailed in Table 1. Although the conductivity anisotropy trend correlates well with the anisotropy ratio measured with polarized Raman in Figure 2e, the contrast in conductivity along the directions of interest is modest, the conductivity along the printing direction being

Table 1. Electrical Conductivity Anisotropic Ratio, Defined as the Conductivity Ratio along the Directions Parallel and Perpendicular to the Brush Stroke, for PEDOT:PSS Films^a

	evaporative regime	mixed regime	Landau–Levich regime
$\sigma_{\parallel}/\sigma_{\perp}$	0.94 ± 0.04	1.23 ± 0.08	1.03 ± 0.03

^aBrush speed of 0.05, 0.5, and 1 mm/s represent the three different regimes.

only ~25% higher than that along the perpendicular direction in the most extreme case corresponding to the mixed regime (0.5 mm/s brushing speed). It is worth noting that although morphological anisotropy was demonstrated for brush-printed PEDOT:PSS before by Wang et al., the authors did not find any correlation with conductivity.¹⁷

Interestingly, during the transition from the mixed regime to the L–L regime (0.8 mm/s), PEDOT exhibited a further redshift in the main Raman peak (Figure 2a) compared to the

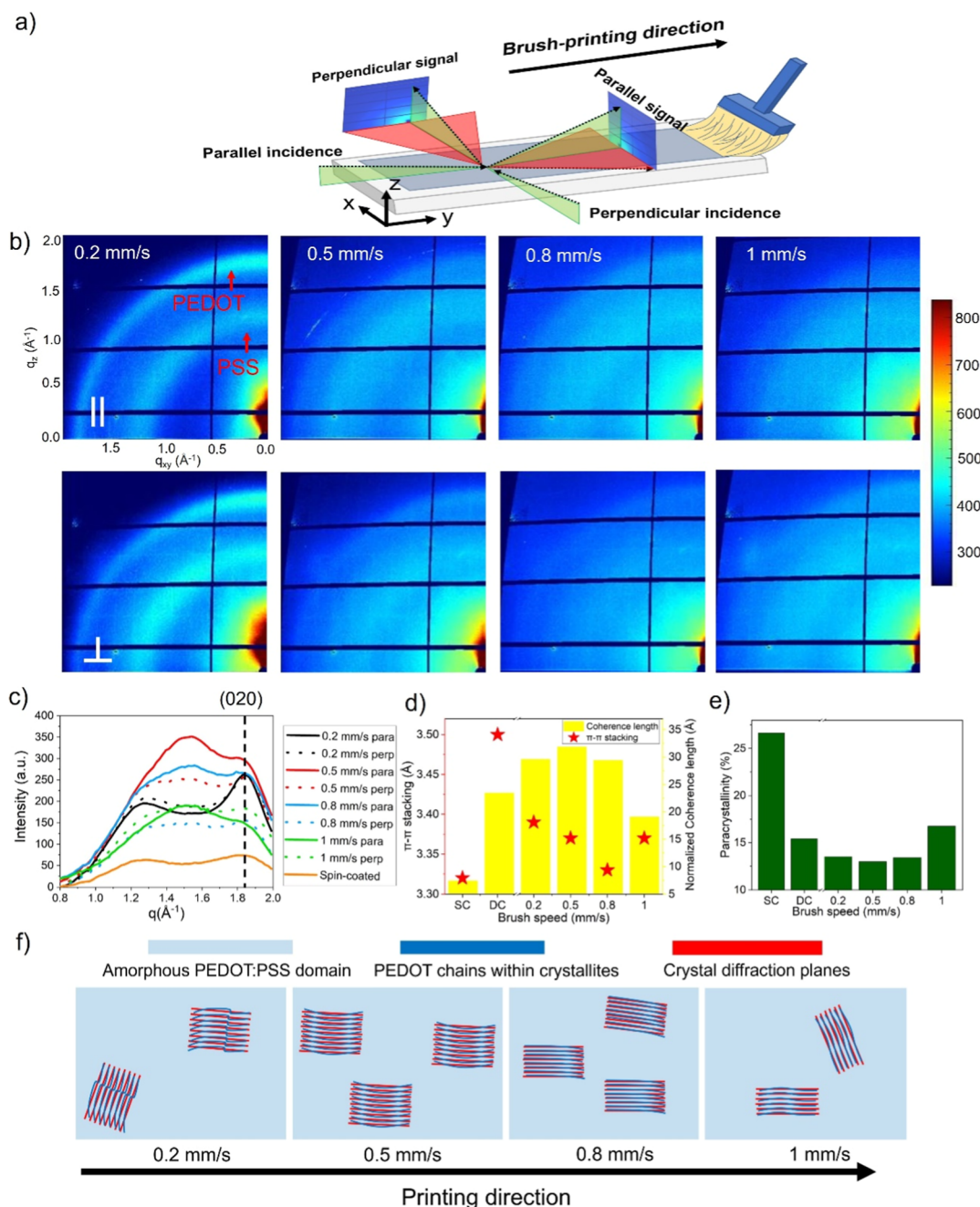


Figure 3. (a) Sketch of the GIWAXS setup and sample placement for measurements along both the parallel (y coordinate) and perpendicular (x coordinate) directions to printing. (b) 2D GIWAXS patterns for films printed at 0.2, 0.5, 0.8, and 1 mm/s in both the parallel (top row) and perpendicular (bottom row) directions to brushing. (c) 1D intensity cut of the GIWAXS patterns in both the parallel and perpendicular directions to brushing (a spin-coated sample is added as a reference). (d) π - π stacking distance and normalized coherence length and (e) paracrystallinity of films printed at different speeds (0.2, 0.5, 0.8, 1 mm/s) and reference spin-coated and drop-casted samples. (f) Schematic depicting the shearing-induced crystalline domain change. The directionality of crystallites is shown in the mixed regime (red, dark blue, and light blue represent the position of the crystal diffraction planes, PEDOT chains within crystallites, and amorphous PEDOT:PSS domains, respectively).

samples brushed at 0.5 mm/s, suggesting that the polymer chains were stretched into their most linear conformation tested. However, the anisotropy ratio was close to 1 (Figure 2e), indicating a loss of molecular alignment. This observation signals that under high-speed brushing the PEDOT molecules are subjected to high shear stress. Hence, they become extended and likely aligned along the brushing direction. However, due to the long meniscus, film consolidation occurs far from the brush tip, and the extended and aligned molecules have time to partially relax before film formation. Because the molecules appear extended but not aligned, the relaxation time for molecule rotation must be necessarily shorter than that for molecule coiling. To investigate the kinetics of the relaxation process, molecular dynamics (MD) simulations were performed (Movie S2) using GAFF2²³ force fields in GROMACS²³ (see Supporting Information for details). The initial system configuration consisted of highly aligned (to the brush direction) straight PEDOT and PSS chains closely packed together. During equilibration (time t_0), the chains reorganized into PEDOT-rich and PEDOT-poor substructures while largely retaining their straight, aligned configuration (Figure 2g). Simulation analysis reveals that PEDOT-rich zones undergo rotation and translation with relative ease, whereas bending is rare and limited due to crystallite resistance (time t_1 , Figure 2h). Those specific scenarios observed in the simulations correlate well with the experimental observations for film brushed at 0.8 mm/s (Figure 2a,e). Notably, within the limited simulation time scale, backbone recoiling was not observed (Movie S2). While the time scales of bending and rotation differ between experiments and simulations, the qualitative insights suggest that polymer chain rotation generally precedes backbone coiling. In summary, brush printing in the mixed regime drives a transformation from isotropic, coiled PEDOT chains into anisotropic, linear chains (Figure 2i), enhancing the electrical conductivity along the printing direction.

Grazing-incidence wide-angle X-ray scattering (GIWAXS) was employed to analyze the material's crystalline structure in the directions parallel and perpendicular to brush printing (Figure 3a).^{24–27} The 2D GIWAXS patterns in Figure 3b show the PEDOT (020) ($q \sim 1.8 \text{ \AA}^{-1}$) and PSS ($q \sim 1.2 \text{ \AA}^{-1}$) diffraction rings. The broad intensity distribution indicates limited long-range crystalline order, typical of PEDOT:PSS. The 1D cuts in Figure 3c were normalized by film thickness after background subtraction to compare differently prepared films (penetration of the X-ray beam through the whole film thickness is assumed²⁸). The following conclusions can be extracted from this analysis: (1) all brush-printed samples exhibited stronger diffraction intensity than the spin-coated sample (Figures S3, 3b,c), indicating a superior degree of crystalline phase; the highest intensity is observed for brushing at 0.5 mm/s. (2) Samples printed at 0.5 and 0.8 mm/s showed stronger signals in the parallel direction than in the perpendicular indicating molecular alignment of the crystalline phase for films printed within the mixed regime; while those at 0.2 and 1 mm/s (evaporative and L–L regimes) display isotropic morphology. The first observation coincides with the findings reported by Wang et al. for brush-printed PEDOT:PSS as “enhanced aggregation”, but the authors did not report anisotropy in the crystalline phase.¹⁷ The molecular alignment along the printing direction at 0.5 mm/s is consistent with the Raman results shown in Figure 2. However, contrary to what was revealed by the Raman study, we also

observed crystalline alignment at 0.8 mm/s. This suggests that it is easier to align the crystallites than the amorphous phase or alternatively that the relaxation time for rotation is longer for large crystallites than for isolated polymer strands. Since PEDOT:PSS is a low crystallinity polymer, the isotropic signal of the predominant amorphous phase in Raman is likely masking the crystallites' alignment at 0.8 mm/s. Overall, a high degree of crystalline phase and molecular alignment in the transport direction should promote transport, aligning with the observation of the highest conductivity observed for a 0.5 mm/s brush speed (Figure 1e). Interestingly, those films brushed at higher speeds also displayed a broad peak located at $q \sim 1.52 \text{ \AA}^{-1}$, between the PEDOT and the PSS rings, which was more prominent in the direction parallel to printing. This finding suggests a shear-induced phase composed of intermixed PEDOT and PSS. To the best of the authors' knowledge, this is the first time this feature has been reported.

To assess quantitatively the impact of crystallinity in electrical transport, peak fitting and paracrystallinity calculations were performed. Figure 3d reveals that the PEDOT π – π stacking distance decreases with increasing coating speed (from 3.39 to 3.33 Å), but this trend reverses at 1 mm/s (3.37 Å). This observation is consistent with the Raman spectroscopy data (Figure 2b), which suggests that extended polymer chains can pack more tightly, resulting in smaller π – π stacking distances. At 1 mm/s, the brushing process enters the L–L regime, and the molecules have time to coil partially (also observed in the Raman blueshift, Figure 2b) before packing in a solid film. Consistently, a drop-casted reference sample, experiencing minimal shear during formation, exhibited a large π – π stacking distance (3.5 Å); and a spin-coated reference sample showed relatively small π – π stacking distances (3.32 Å), likely due to the strong shear forces exerted by the centrifugal force during spinning. Smaller π – π stacking distances are desired as they promote better intramolecular charge transport. Nevertheless, the π – π stacking distance alone does not provide sufficient information to correlate crystal packing and transport.

To complete the study, a normalized (see Materials and Methods) coherence length of the PEDOT crystallites was estimated for films brushed at different speeds and for spin-coated and drop-casted reference samples (Figure 3d), allowing for the calculation of the paracrystallinity of the films (Figure 3e). Paracrystallinity, defined as the square root of full width at half maximum (fwhm) over peak position (see Materials and Methods) is inversely proportional to the square root of the number of molecules in an ideal crystallite, quantifying the extent of disorder in the crystalline domains. Hence, paracrystallinity has been claimed to be a reliable indicator to relate crystal packing with mobility.^{25,29} Specifically, the lower the paracrystallinity, the better the mobility. Figure 3e reveals that brush-printed (13–16.7%) and drop-casting films (15.4%) presented lower paracrystallinity than that of spin-coated materials (26.6%). The high paracrystallinity of spin-coated films could be explained by the drying process of the technique: as the solvent dries uniformly across the whole film surface, multiple nucleation events occur simultaneously. Crystal growth would be then hindered by neighboring nuclei, leading to a highly polycrystalline structure composed of many small crystallites. On the other hand, the directional drying of drop-casting films (from the edge to the center), and especially of brush-printed films (unidirectional), favors orderly molecular assembly promoting crystal growth

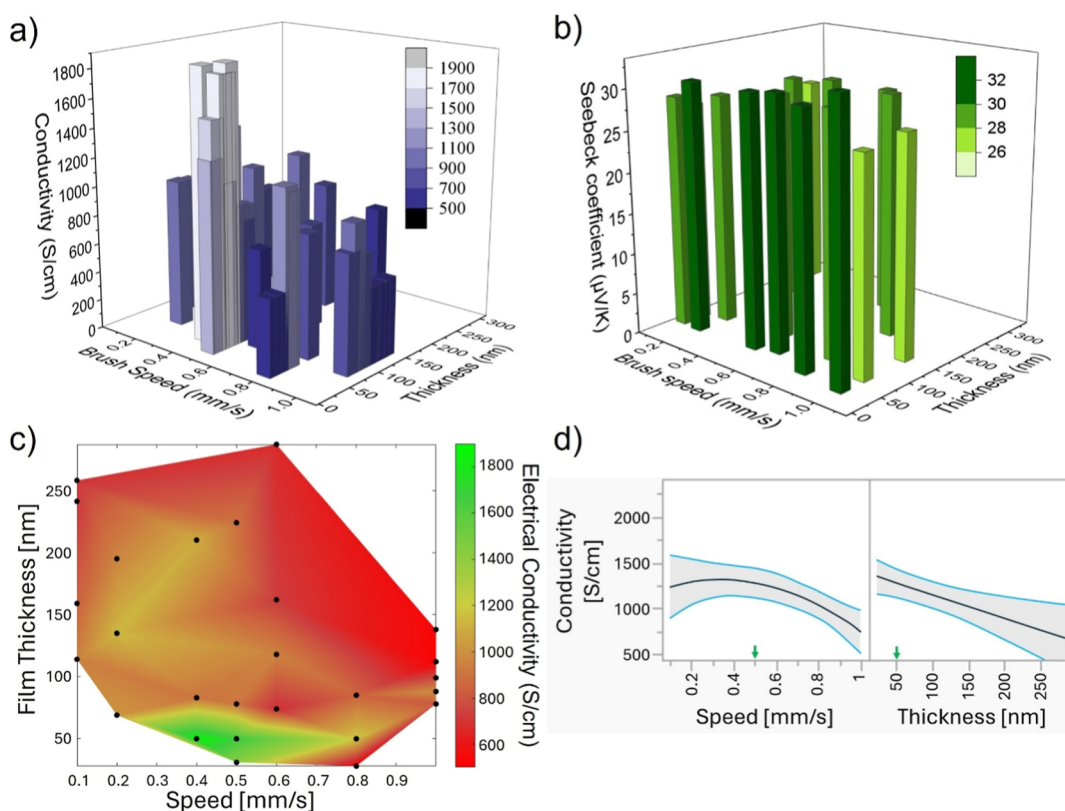


Figure 4. 3-Way relationship between printing speed, film thickness, and (a) electrical conductivity and (b) Seebeck coefficient. (c) 2D projected color map plot by linear interpolation of (a). (d) Deconvoluted influence of brush speed and thickness on the electrical conductivity. For comparison, the conductivity of a spin-coated film is ~ 600 S/cm (thickness ~ 70 nm) and the conductivity of a drop-casting film ~ 400 S/cm (thickness ~ 1 μm).

and propagation, leading to fewer but larger crystallites. Within the brush-printed samples, those produced in the mixed regime (0.5 mm/s) presented the lowest paracrystallinity of 13%, suggesting superior crystal order induced by controlled unidirectional solution shearing at the meniscus drying front. This result supports the highest conductivity observed for the 0.5 mm/s brush speed (Figure 1e). Consistently with Raman results in Figure 2, at 1 mm/s (L–L regime), the drying process resembled the one of drop-casting, driving an increase in paracrystallinity to values approaching those of drop-casted film. A brush speed of 0.8 mm/s leads to interesting results, as it yields the most extended molecules (Figure 2a) and shorter π – π stacking distance (Figure 3d) but not the lowest paracrystallinity. This observation suggests that this speed marks the onset of molecule relaxation, and in agreement with the conclusions from the MD analysis in Figure 2g,h, molecules might rotate and hinder bundling into aggregates before film consolidation. We have not found any available literature that directly applies the concept of paracrystallinity to the PEDOT:PSS system. PEDOT:PSS has a complicated phase-separated structure, but we observed a strong correlation between paracrystallinity and electrical conductivity especially for those samples brushed at 0.5, 0.8, and 1 mm/s, as depicted in Figure S4. The highly correlated samples have the commonality of containing the most extended and oriented backbones (in both crystalline and amorphous phases) according to the Raman study in Figure 2a,e. Molecules in the amorphous phase with extended and oriented backbones are expected to act as highly effective intercrystallite tie chains leading to materials where the conductivity bottleneck is likely

the intracrystallite mobility.³⁰ Hence, changes in crystallinity must be directly reflected in conductivity, as is the case in Figure S4. This analysis hints at the fact that a direct paracrystallinity-mobility relationship exists in PEDOT:PSS crystallites, but this relationship is observable only for samples in which crystallites are strongly interconnected. However, further studies are needed to validate this “paracrystallinity-controlled conductivity”, especially in anisotropic systems where mobility might be also influenced by molecular alignment. Although low paracrystallinity is just one of many contributors to conductivity (being that molecular alignment, degree of crystallinity, and the morphology of the amorphous phase are also important factors), the trend found in paracrystallinity aligns well with the trend in conductivity depicted in Figure 1e for the mixed regime, suggesting the important role of paracrystallinity in the conductivity of PEDOT:PSS.

Given our GIWAXS analysis, we propose a description of the morphology evolution for the crystalline phase of PEDOT:PSS films upon brush-printing as follows (Figure 3f): The total amount of crystal phase increases by brush printing compared to a spin-coated sample, with 0.5 mm/s being the optimal brushing speed. Only the crystalline domains of films brushed at the intermediate speeds corresponding to the mixed regime (0.5, 0.8 mm/s) show alignment in the printing direction. As brush speed increases within the mixed regime, the brush-induced shear extends the PEDOT molecules, promoting tight packing, which manifests as a reduction of the π – π stacking distance. At the same time, brush printing promotes the orderly packing of molecules

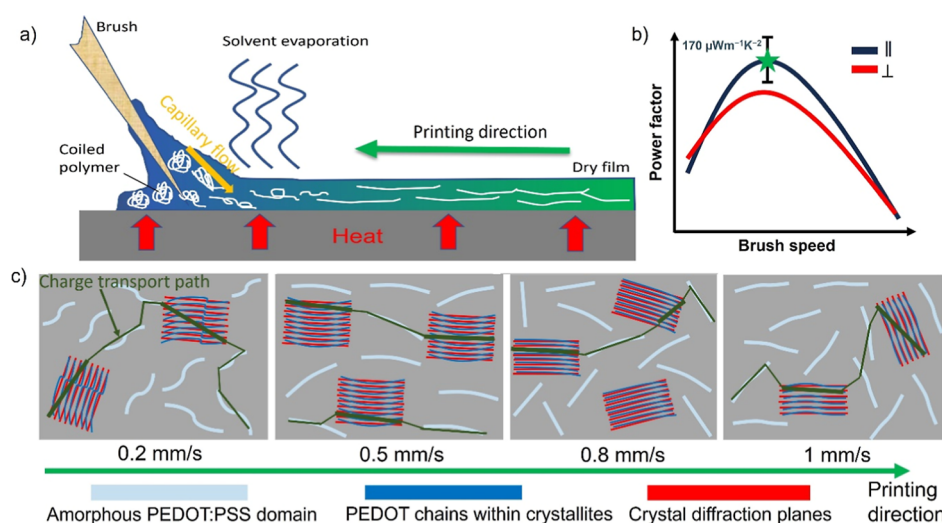


Figure 5. (a) Sketch of stretching polymer chains under the effect of brush printing (b) Power factor change along the parallel and perpendicular direction to printing under different brush speeds. (c) Schematic depicting the full morphology (both crystalline and amorphous phase) of PEDOT:PSS films brushed at different speeds (red, dark blue, and light blue represent the position of the crystal diffraction planes, PEDOT chains within crystallites, and amorphous PEDOT:PSS domains; dark green indicates likely charge transport paths, with thicker lines representing easier transport).

likely by preorienting them and bundling them, which leads to an increase in coherence length (represented by large crystallites in Figure 3f), and a reduction in paracrystallinity (represented by a higher number of molecules in each crystallite), which are optimal at 0.5 mm/s brushing. When entering the L–L regime, even though the shear force should be the highest, the fact that the film dries far from the meniscus makes the sheared molecules rotate and partially coil back before assembling in the bundles, reducing the π – π stacking distance and the coherence length and increasing paracrystallinity.

Figure 1e shows the dependence of conductivity on brush printing regimes associated with a certain brush speed and film thickness. By performing different material characterization experiments for films with similar thicknesses, we have concluded that the dependence of conductivity on brush speed is underpinned by a morphological change induced by shear flow. However, the potential effect of film thickness encoded in the conductivity trend depicted in Figure 1e must be also extracted to find the maximum achievable conductivity of the material. To isolate the effects of speed and thickness, we adjusted the amount of ink fed to the brush, achieving a range of film thicknesses for each brush speed. Figure 4a illustrates the 3-way relationship between electrical conductivity, printing speed, and film thickness. By individually controlling the brush speed and film thickness, the electrical conductivity could be increased from 600 (spin-coated) to a maximum of 1900 S/cm. The Seebeck coefficient had less than 10% variation compared to the spin-coated sample and remained largely unaffected by changes in thickness and speed (Figure 4b). 3D scattering plots with error bars are shown in Figure S5. The color map of Figure 4c visualizes the change in electrical conductivity with speed and thickness, where higher electrical conductivities were achieved for a relatively low film thickness and intermediate brushing speeds. Statistics analysis was used to deconvolute the significant influence of brush speed and film thickness on conductivity (with p-values below 0.05, Table S1), as shown in Figure 4d. This study revealed significant effects for both the thickness

and brush speed. The quadratic effect observed for brush speed with conductivity implies a maximum conductivity at medium speed (in the mixed regime), confirming the trend highlighted in Figure 1b regardless of the thickness. Moreover, this quadratic behavior is accompanied by a linear increase in conductivity for reducing the thickness. The explanation for this thickness dependence is 2-fold: (1) the average shear stress in the wet film during casting (assuming a Couette flow) is proportional to *brush speed/meniscus thickness*, considering the definition of a fluid shear stress $\tau = \mu \partial v_x / \partial y$, with μ being the dynamic viscosity, v_x the flow velocity, and y the direction along the thickness.¹⁶ This relation explains why conductivity increases with brush printing (before the L–L regime) and for thin films (that arise from a thin meniscus). (2) Enhanced conductivity for thin films (below a few 10 s of nm in thickness) has also been attributed to stronger in-plane polymer chain confinement promoting in-plane charge transport.^{31,32} However, the high roughness of brush-printed films imposes 50 nm as the minimum thickness to achieve uniform films (Figure S8 and Table S2). The electrical conductivity reaches a maximum of 1900 ± 300 S/cm at a brush speed of 0.5 mm/s for a small film thickness of 50 nm. This maximum marks a significant enhancement of 86% compared to a sample printed at a lower speed of 0.2 mm/s (1019 S/cm) with a similar small thickness or a 76% enhancement compared to a thicker 125 nm thick film printed at the same speed of 0.5 mm/s (1078 S/cm). For comparison, a standard spin-coated film displayed 600 S/cm (~ 70 nm thick) and drop-casted film 400 S/cm (~ 1 μm thick). To the best of our knowledge, 1900 S/cm is the highest conductivity reported for PEDOT:PSS not subjected to postprocessing by dipping in acid or polar solvents.

DISCUSSION

Brush printing, a uniaxial meniscus-guided solution deposition technique, induces shear stress on the ink located in the meniscus area during solvent drying. This shear stress leads to changes in the final dry film morphology forming at the meniscus if the solvent dries fast enough to lock in place the

molecules in the solid film. In the brush printing setup (Figure 5a), the brush speed regulates the shear stress and the substrate temperature controls the solvent evaporation rate, allowing access to different material morphologies and properties. We obtain the highest thermoelectric power factor for PEDOT:PSS (Figure 5b) in the mixed regime of brush-printing, where polymer chains exhibit significant extension, uniaxial alignment, and packing order, inducing anisotropy and facilitating charge transport along the printing direction. Studies on relaxation from the boundary between the mixed and the L–L regime reveal that polymer chain rotation reduces anisotropy before recoiling. This is an important finding applicable to other meniscus-guided printing techniques.

In terms of the crystalline phase, brush-printed samples show a higher degree of crystal packing than spin-coated samples, and crystalline orientation along the printing direction is found in films brushed at 0.5 and 0.8 mm/s. The fact that films brushed at 0.8 mm/s look isotropic in polarized Raman spectroscopy but not in GIWAXS suggests that the crystalline phase is preferentially aligned but the amorphous part is not. Then the film's low crystallinity results in the majority isotropic amorphous phase signal masking the alignment of the minority crystalline phase in the Raman signal. Films brushed at 0.5 mm/s display anisotropy in both polarized Raman spectroscopy and GIWAXS. One may think that the higher degree of crystallinity of films brushed at 0.5 mm/s compared to their counterparts brushed at 0.8 mm/s could contribute to making the crystalline aligned phase noticeable in the polarized Raman spectrum. However, in a generally low-crystallinity material like PEDOT:PSS, the crystallinity contribution should not be strong enough to justify the high anisotropy found in Raman, which suggests that at 0.5 mm/s brush speed the amorphous phase is also preferentially aligned along the printing direction. The more aligned the amorphous phase, the straighter the tie chains connecting crystallites and the better the intercrystallite transport, leading to an "intracrystallite-limited" conductivity regime for moderate brush printing speeds in which conductivity is linearly correlated with paracrystallinity (Figure S4). Furthermore, brush-printed samples show reduced paracrystallinity compared to spin-coated samples, which contributes to enhancing charge transport within crystalline domains. Intermediate speeds within the mixed regime optimize the combination of a short π – π stacking distance and large coherence length, leading to the lowest paracrystallinity and the best expected intracrystallite transport. Improved intracrystallite transport enhances conductivity in both parallel and perpendicular directions, leading to a 216% increase in electrical conductivity along the printing direction but only a 23% difference in the anisotropic ratio. Finally, an interesting new phase attributed to intermixed PEDOT and PSS arises at medium brushing speeds. It is not clear what the role of this new phase is in transport, and its clarification deserves further study.

The superior intermolecular packing of PEDOT:PSS brushed at 0.5 mm/s, manifested by a high degree of crystallinity and a low paracrystallinity, combined with the preferential molecular alignment along the printing direction encountered for both the amorphous and crystalline phases (Figure 5c), can explain why the 0.5 mm/s brush speed yielded the highest conductivity in this study (1900 ± 300 S/cm). Meanwhile, the Seebeck coefficient remained consistent across different printing conditions. The Seebeck coefficient is very sensitive to doping levels, which are not modified in this study.

Hence, a maximum power factor of $170 \pm 37 \mu\text{W m}^{-1} \text{K}^{-2}$ was achieved for a 0.5 mm/s brush speed. These values of conductivity and power factor are the highest reported for PEDOT:PSS without posttreatment.

Beyond these findings, this study highlights brush printing as a versatile and scalable tool for the morphology control and patterning of solution-processed thermoelectric polymers. To further improve performance, future efforts should focus on enhancing the extended polymer chains while locking in place chain alignment to increase crystallinity and anisotropy, which is not possible now at the boundary between mixed and L–L regimes. For instance, modifying the solution viscosity or printing on rough substrates could be valid strategies to increase shear stress without increasing the brush speed. Besides boosting conductivity, further increasing crystallinity may potentially improve the Seebeck coefficient, contributing to enhanced thermoelectric performance.³³ Brush printing can be applied to both p- and n-type polymers, broadening its utility for thermoelectric applications. Combined with flexible substrates, brush printing offers a cost-effective pathway to all-printed, flexible thermoelectric generators, showcasing the significant potential for real-world applications. Furthermore, the systematic exploration of the link among process, morphology, and performance provides a framework for optimizing not only thermoelectric materials but also functional polymers for other applications.

CONCLUSIONS

This study demonstrates that brush printing is an effective and scalable technique for enhancing the thermoelectric performance of PEDOT:PSS films by tuning their morphology. Through controlled shear stress and solvent evaporation, brush printing induces molecular extension, uniaxial alignment, and improved crystalline packing, leading to a significant increase in electrical conductivity (1900 ± 300 S/cm) and power factor ($170 \pm 37 \mu\text{W m}^{-1} \text{K}^{-2}$) without the need for posttreatment. The optimized morphology, particularly at a brush speed of 0.5 mm/s, enhances charge transport by aligning both crystalline and amorphous phases, reducing paracrystallinity and improving intracrystallite transport. These findings establish brush printing as a promising and simple approach for producing high-performance, patterned thermoelectric films suitable for TEG applications. The versatility of brush printing in processing conjugated polymers highlights its potential for advancing cost-effective, flexible thermoelectric generators, and other functional polymer applications.

MATERIALS AND METHODS

Materials. PEDOT:PSS (Clevios PH1000) was purchased from Heraeus Electronic Materials. Dimethyl sulfoxide (DMSO) was purchased from Sigma-Aldrich (Overijse, Belgium). Ethanol, acetone, and 2-propanol were purchased from Chem-Lab (Zedelgem, Belgium). All of the materials were used as purchased.

Preparation of Films. The aqueous PEDOT:PSS dispersion was mixed with 5 vol% DMSO and stirred for 2 h, followed by filtration through 0.45 μm PET filters. PEDOT:PSS films were deposited on SiO_2 -coated (300 nm) silicon wafers and glass substrates by brushing. Prior to printing, the substrates were cleaned sequentially with acetone, 2-propanol, and water by submerging them in an ultrasonic bath. To improve wettability, ozone plasma treatment was applied to the substrates for 30 min before printing. The brush (Zhu Jiang) was sonicated in ethanol and water for 30 min to remove contaminants and then dried on a vacuum hot plate at 50 °C for 2 h. The brush's movement was precisely controlled by mounting it on a 3D printer equipped with a Zaber actuator, which was capable of regulating substrate temperature and coating speed. Films were deposited on a 100 °C heated substrate with the brush tip in slight contact with the substrate (the position was controlled by the 3D printer motor). After

printing, the samples were annealed at 120 °C for 1 h in ambient atmosphere.

GIWAXS. GIWAXS measurements were conducted at BM26 (DUBBLE) at the European Synchrotron Radiation Facility (ESRF). Samples were exposed to 12 keV X-rays for 30 s. The sample-to-detector distance (347 mm) was calibrated, and the grazing incidence angle was set to 0.12°. A helium fly tube was positioned between the sample and detector to minimize air scattering. 2D GIWAXS images were captured by using a Pilatus 1 M detector and visualized with GIXSAXS, a tool developed by DUBBLE@ESRF. The GIXSGUI software was used to obtain data for each sample and the Si wafer background. The GIWAXS images of brush-printed PEDOT:PSS samples were reshaped into q_z vs q_y instead of q_z vs q_x , which is typically found in the literature. The typical reshaping method is based on the assumption that the sample is in-plane isotropic; however, the brush-printed PEDOT:PSS films were found to be in-plane anisotropic in UV-vis and Raman spectra. Therefore, our GIWAXS images were reshaped into q_z vs q_y , and the reshaped image was in a dome-like shape instead of a butterfly like shape.³⁴ The 1D line-cuts of the samples were then processed by subtracting the background signal and normalizing the intensity by the film thickness.

To set the π - π stacking peak position in the GIWAXS pattern, we programmed a MATLAB script that executes the following steps: first, it assigns an estimated q value and width to the following Gaussian peaks (in red in Figure S6): lamella (100) and (200), PSS, and PEDOT π - π stacking. Moreover, a broad Gaussian peak is introduced to represent the background signal (Figure S6). Then an iterative fitting loop is run to fit simultaneously all of the different peaks until an optimized fit is found. From the optimized fitting results, we can precisely determine the π - π stacking peak position and full-width at half-maximum (fwhm) and calculate the corresponding stacking distance. Particularly the GIWAXS 1D line-cut is further fitted with the Gaussian function by Matlab's "lsqcurvefit" function. The π - π packing distance is calculated by converting the PEDOT (020) peak position from reciprocal into real space. With the fwhm of the PEDOT (020) peak, Δq_{020} , the π - π coherence length is calculated by the Scherrer equation, shown as follows

$$L_c = \frac{2\pi K}{\Delta q_{020}} \quad (1)$$

where K is a constant related to crystal shape arbitrarily considered as 1 in this research because the interest lies in the trend of coherence length rather than in its absolute value. Hence, we call the coherence length the "normalized coherence length". The paracrystallinity of the π - π direction is also calculated from the fwhm of the PEDOT (020) peak with the following equation

$$g_{\pi-\pi}(\%) = \sqrt{\frac{\Delta q_{020}}{2\pi \times q_{020}}} \quad (2)$$

where q_{020} and Δq_{020} are the position and fwhm of the (020) peak, respectively.^{35,36} To compare the amount of crystalline phase among different samples, we ensured that all samples (2 × 2 mm) were subjected to the same X-ray beam intensity, exposure time, and incident angle. For data processing, the GIWAXS 1D intensity linecuts were first background-subtracted and then normalized by film thickness (to equalize the overlapping volume between the X-ray beam and material across samples). Under these conditions, the integrated area under the 1D linecuts provides a relative measure of the amount of crystalline phase present in the sample.

Raman Spectroscopy. The presented Raman spectra were collected using a Horiba LabRAM HR Evolution. Raman shift was calibrated with an NIST-certified Horiba SP-RCO calibration objective. The samples were probed with a 1.7 W 633 nm HeNe laser with a 20 s exposure time. For each condition, three spectra were collected from different points on the same sample and averaged. Background subtraction was performed using the LABRAM HR

Evolution system, followed by normalization of the main peak $C_\alpha = C_\beta$ intensity to facilitate a peak shift comparison.

Molecular Dynamics Simulation. PEDOT is represented in its bipolar form, with chain lengths of 6 (+2 charge) and 12 (+4 charge). PSS is represented by chains of length averaging around 46, with a mixture of protonated and deprotonated sulfonate group. Initial configurations (t_0) of the simulations feature a 1:2.5 ratio of PEDOT:PSS chains, solvated in an aqueous medium with 5 vol% of DMSO.

UV-Visible Spectroscopy. A UV-Vis spectrometer (UV1601, Shimadzu) equipped with a polarizer (Shimadzu ASSY, wavelength range 400–800 nm) was used to measure absorbance in the parallel and perpendicular orientations relative to the printing direction. The polarizer operates within a wavelength range of 400–800 nm. Polarized UV-Vis spectroscopy is a valuable technique for characterizing the anisotropy of materials, providing insight into how light absorption varies with the polarization direction. To achieve this, a linear polarizer is placed before the sample to control the polarization of the incident light. When the polarization direction aligns with the molecular backbone, the interaction between light and the molecules is maximized, resulting in higher absorbance. Conversely, when the polarization direction is perpendicular to the molecular alignment, the interaction is weaker, leading to a lower absorbance. This difference in absorption allows polarized UV-Vis spectroscopy to effectively reveal the molecular alignment within the material.

Thermoelectric Characterization. Conductivity: Conductive silver ink (DM-SIP-2006, DYCOTEC) was applied to the printed samples as electrodes for 4-point measurements. Electrical resistance was measured at room temperature using a Keithley 2000 multimeter. Lengths were measured with a caliper, and a Dektak XT profilometer was used to scan the cross-sectional profiles of the printed films (cross-sectional profiles in Figure S7). Each parameter was measured three times to obtain an average value for accuracy.

Seebeck Coefficient: The Seebeck coefficient was measured using a custom-built setup. This setup included two Peltier units, with the temperature difference controlled by adjusting the current source. To avoid the cold finger effect, the same probes were used to measure both the temperature difference and the resulting thermal voltages. For each sample, voltage measurements were taken at five different temperature differences and a linear fit of these points was used to determine the relative Seebeck coefficient as the line slope. To account for the Seebeck coefficient of the probes, the setup was calibrated using a constantan reference sample. A correction factor of +7 $\mu\text{V/K}$ was added to the relative Seebeck coefficient to obtain the absolute value.

Statistics. The anisotropy ratio in Figure 2e is presented by using the geometric mean as the central value, with geometric error represented by the 25th and 75th percentiles, capturing the asymmetric distribution of the data. For all other data sets, the standard error (calculated as the standard deviation divided by the square root of the number of measurements) was used to represent the error. Each reported average is based on at least three data points, with the standard error reflecting measurement variability. Power factor errors were calculated using standard propagation theory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c00803>.

POM of a brush-printed PEDOT:PSS film; AFM phase images of drop-casting and brush-printed films; 2D GIWAXS pattern of a spin-coated PEDOT:PSS film; correlation between electrical conductivity and paracrystallinity of brush-printed PEDOT:PSS films; 3-way relationship with error bars between printing speed, film thickness and electrical conductivity and Seebeck coefficient; GIWAXS 1D linecut with peak fitting before and after background subtraction; cross-section profiles

of PEDOT:PSS lines; statistics study on thickness influence on electrical conductivity; and computational details (PDF)

Meniscus movement during brush printing (AVI)

MD simulation on relaxation of stretched PEDOT:PSS chains (MOV)

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Notes

The authors declare no competing financial interest.

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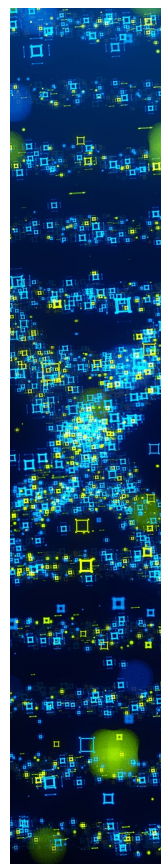
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Due to a production error, this paper was published ASAP on March 21, 2025, with an error in the Thermoelectric Characterization section. The corrected version was reposted on March 21, 2025.



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